Assignment #4

The zip file of `Files4A4.zip` contains three java files and two `.wav` files.

1. Within `codes**8490`, create two new java files `A4Shapes**.java` and `Assignment4**.java`, where `**` is your initials.

2. From the Blender file provided in the previous assignment, extract a component for the switch button and save it to the same folder of the other .obj files.

3. Code your `A4Shapes**.java` and `Assignment4**.java` files to render the virtual world of a desktop fan as shown in the figures below.

   (a) Add two switch buttons to the switch box, one at each side, with color being set to red.

   (b) Make the left switch button responding to the pressing of `z` key and the right button to `x` key with a color change between red and green. While red signals `ON`, green signals `OFF`.

   Hint: The `Switch` class allows the selective display of child objects, as discussed in Slides `J2.S43-44`.

   (c) Set the switch button on the left as the pause key and the button on the right as the power key (as in Assignment #3) so that the desktop fan oscillates and/or rotates depending on the on/off status of switch buttons.

   (d) Make the two switch buttons also responding to mouse clicking in the same way as they respond to the `z` and `x` keys.

   Hint: While `Code4Assign4.java` shows how to change the color of an object via mouse clicking, the same result can be achieved by alternating the child objects in a `Switch` group.

   (e) Play a wind sound of your choice whenever the blades rotate.

   Hint: The demo program of `SoundPointJOAL.java` works with `SoundUtilityJOAL.java` to produce an ocean sound in the background and the cow sound at a point.

   (f) Update the string label on the base to identify your work, and make your program easy to comprehend by adding adequate amount of comments.

4. Locate the folder that contains all files of your project `Comp8490`; produce a zip file of your project folder; and submit it online before due.